

Longikaurin A induces ferroptosis and inhibits glioblastoma progression through DNA methylation - Mediated GPX4 suppression.

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Glioblastoma (GBM) is the most common primary intracranial tumor highly resistant to conventional clinical chemotherapy. Recently, the induction of ferroptosis is emerging as a putative strategy to treat various tumors. However, the identification of the effective and applicable tumor ferroptosis-inducing agents remains challenging. In this study, we showed that longikaurin A (LK-A), a natural diterpenoid isolated from the medicinal plant *Isodon ternifolius* with strong anti-GBM capacities, induced remarkable GBM cell ferroptosis along with suppressing the key anti-ferroptosis factor glutathione peroxidase 4 (GPX4). GPX4 promoter contains conserved CpG islands. The LK-A-induced GPX4 suppression coincided with the inhibition of ten-eleven translocation 2 (TET2), a key DNA demethylation enzyme and an increase in the hypermethylation of the GPX4 promoter. Further, LK-A promoted the GBM ferroptotic alterations and inhibited GBM progression in both subcutaneous and orthotopic xenograft mouse models, whereas GPX4 overexpression largely abrogated its anti-GBM effects both in vitro and in vivo, suggesting that LK-A inductions of the DNA methylation-incurred GPX4 suppression and ferroptosis are crucial for its anti-GBM functions. Together, our study has elaborated an important epigenetic pathway of GBM ferroptosis and uncovered a critical pharmacological property of LK-A for treating GBM patients.

Understanding the role of ten-eleven translocation family proteins in kidney diseases.

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Epigenetic mechanisms play a critical role in the pathogenesis of human diseases including kidney disorders. As the erasers of DNA methylation, Ten-eleven translocation (TET) family proteins can oxidize 5-methylcytosine (5mC) to 5-hydroxymethylcytosine (5hmC), 5-formylcytosine (5fC), and 5-carboxylcytosine (5caC), thus leading to passive or active DNA demethylation. Similarly, TET family proteins can also catalyze the same reaction on RNA. In addition, TET family proteins can also regulate chromatin structure and gene expression in a catalytic activity-independent manner through recruiting the SIN3A/HDAC co-repressor complex. In 2012, we reported for the first time that the genomic 5-hydroxymethylcytosine level and the mRNA levels of Tet1 and Tet2 were significantly downregulated in murine kidneys upon ischemia and reperfusion injury. Since then, accumulating evidences have eventually established an indispensable role of TET family proteins in not only acute kidney injury but also chronic kidney disease. In this review, we summarize the upstream regulatory mechanisms and the pathophysiological role of TET family proteins in major types of kidney diseases and discuss their potential values in clinical diagnosis and treatment.

Functions of intrinsic disorder in proteins involved in DNA demethylation during pre-implantation embryonic development.

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DNA demethylation is involved in many biological processes during pre-implantation embryonic development in mammals. To date, the complicated mechanism of DNA demethylation is still not fully understood. Ten-eleven translocation family (TET3, TET1 and TET2), thymine DNA glycosylase (TDG) and

DNA methyltransferase 1 (DNMT1) are considered the major protein enzymes of DNA demethylation in pre-implantation embryos. TET3, TET1, TET2, TDG, and DNMT1 contain abundant levels of intrinsically disordered protein regions (IDPRs), which contribute to increasing the functional diversity of proteins. Thus we tried to explore the complicated DNA demethylation in pre-implantation embryos from the intrinsic disorder perspective. These five biological macromolecules all have DNA demethylation-related functional domains. They can work together to fulfill DNA demethylation in pre-implantation embryos through complex protein-protein interaction networks. Intrinsic disorder analysis results showed these proteins were partial intrinsically disordered proteins. Many identifiable disorder-based DNA-binding sites, protein-binding sites and post-translational modification sites located in the intrinsically disordered regions, and DNA demethylation deficiency point mutations in the IDPRs could significantly change the local disorder propensity of these proteins. To the best of our knowledge, this work provides a new viewpoint for studying the mechanism of DNA methylation reprogramming during mammalian pre-implantation embryonic development.

Cytosine-Based TET Enzyme Inhibitors.

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DNA methylation is known as the prima donna epigenetic mark for its critical role in regulating local gene transcription. Changes in the landscape of DNA methylation across the genome occur during cellular transition, such as differentiation and altered neuronal plasticity, and become dysregulated in disease states such as cancer. The TET family of enzymes is known to be responsible for catalyzing the reverse process that is DNA demethylation by recognizing 5-methylcytosine and oxidizing the methyl group via an Fe(II)/alpha-ketoglutarate-dependent mechanism. Here, we describe the design, synthesis, and evaluation of novel cytosine-based TET enzyme inhibitors, a class of small molecule probes previously underdeveloped but broadly desired in the field of epigenetics. We identify a promising cytosine-based lead compound, Bobcat339, that has mid-μM inhibitor activity against TET1 and TET2, but does not inhibit the DNA methyltransferase, DNMT3a. In silico modeling of the TET enzyme active site is used to rationalize the activity of Bobcat339 and other cytosine-based inhibitors. These new molecular tools will be useful to the field of epigenetics and serve as a starting point for new therapeutics that target DNA methylation and gene transcription.

APOBEC3A efficiently deaminates methylated, but not TET-oxidized, cytosine bases in DNA.

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AID/APOBEC family enzymes are best known for deaminating cytosine bases to uracil in single-stranded DNA, with characteristic sequence preferences that can produce mutational signatures in targets such as retroviral and cancer cell genomes. These deaminases have also been proposed to function in DNA demethylation via deamination of either 5-methylcytosine (mC) or TET-oxidized mC bases (ox-mCs), which include 5-hydroxymethylcytosine, 5-formylcytosine and 5-carboxylcytosine. One specific family member, APOBEC3A (A3A), has been shown to readily deaminate mC, raising the prospect of broader activity on ox-mCs. To investigate this claim, we developed a novel assay that allows for parallel profiling of activity on all modified cytosines. Our steady-state kinetic analysis reveals that A3A discriminates

against all ox-mCs by >3700-fold, arguing that ox-mC deamination does not contribute substantially to demethylation. A3A is, by contrast, highly proficient at C/mC deamination. Under conditions of excess enzyme, C/mC bases can be deaminated to completion in long DNA segments, regardless of sequence context. Interestingly, under limiting A3A, the sequence preferences observed with targeting unmodified cytosine are further exaggerated when deaminating mC. Our study informs how methylation, oxidation, and deamination can interplay in the genome and suggests A3A's potential utility as a biotechnological tool to discriminate between cytosine modification states.

Tet1 and Tet2 Protect DNA Methylation Canyons against Hypermethylation.

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DNA methylation is a dynamic epigenetic modification with an important role in cell fate specification and reprogramming. The Ten eleven translocation (Tet) family of enzymes converts 5-methylcytosine to 5-hydroxymethylcytosine, which promotes passive DNA demethylation and functions as an intermediate in an active DNA demethylation process. Tet1/Tet2 double-knockout mice are characterized by developmental defects and epigenetic instability, suggesting a requirement for Tet-mediated DNA demethylation for the proper regulation of gene expression during differentiation. Here, we used whole-genome bisulfite and transcriptome sequencing to characterize the underlying mechanisms. Our results uncover the hypermethylation of DNA methylation canyons as the genomic key feature of Tet1/Tet2 double-knockout mouse embryonic fibroblasts. Canyon hypermethylation coincided with disturbed regulation of associated genes, suggesting a mechanistic explanation for the observed Tet-dependent differentiation defects. Based on these results, we propose an important regulatory role of Tet-dependent DNA demethylation for the maintenance of DNA methylation canyons, which prevents invasive DNA methylation and allows functional regulation of canyon-associated genes.

TET2 in Normal and Malignant Hematopoiesis.

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The ten-eleven translocation (TET) family of enzymes were originally cloned from the translocation breakpoint of t(10;11) in infant acute myeloid leukemia (AML) with subsequent genomic analyses revealing somatic mutations and suppressed expression of TET family members across a range of malignancies, particularly enriched in hematological neoplasms. The TET family of enzymes is responsible for the hydroxylation of 5-methylcytosines (5-mC) to 5-hydroxymethylcytosine (5-hmC), followed by active and passive mechanisms leading to DNA demethylation. Given the complexity and importance of DNA methylation events in cellular proliferation and differentiation, it comes as no surprise that the TET family of enzymes is intricately regulated by both small molecules and regulatory cooperating proteins. Here, we review the structure and function of TET2, its interactions with cooperating mutations and small molecules, and its role in aberrant hematopoiesis.

Substrate DNA length regulates the activity of TET 5-methylcytosine dioxygenases.

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The ten-eleven translocation (TET) isoforms (TET1-3) play critical roles in epigenetic transcription regulation. In addition, mutations in the TET2 gene are frequently detected in patients with glioma and myeloid malignancies. TET isoforms can oxidize 5-methylcytosine to 5-hydroxymethylcytosine, 5-formylcytosine, and 5-carboxylcytosine, by iterative oxidation. The in vivo DNA demethylation activity of TET isoforms may depend on many factors including enzyme's structural features, its interaction with DNA-binding proteins, chromatin context, DNA sequence, DNA length, and configuration. The rationale for this study is to identify the preferred DNA length and configuration in the substrates of TET isoforms. We have used a highly sensitive LC-MS/MS-based method to compare the substrate preference of TET isoforms. To this end, four DNA substrate sets (S1, S2, S3, S4) of different sequences were chosen. In addition, in each set, four different lengths of DNA substrates comprising 7-, 13-, 19-, and 25-mer nucleotides were synthesized. Each DNA substrate was further used in three different configurations, that is, double stranded symmetrically-methylated, double stranded hemi-methylated, and single stranded single-methylated to evaluate their effect on TET-mediated 5mC oxidation. We demonstrate that mouse TET1 (mTET1) and human TET2 (hTET2) have highest preference for 13-mer dsDNA substrates. Increasing or decreasing the length of dsDNA substrate reduces product formation. In contrast to their dsDNA counterparts, the length of ssDNA substrates did not have a predictable effect on 5mC oxidation. Finally, we show that substrate specificity of TET isoforms correlates with their DNA binding efficiency. Our results demonstrate that mTET1 and hTET2 prefer 13-mer dsDNA as a substrate over ssDNA. These results may help elucidate novel properties of TET-mediated 5mC oxidation and help develop novel diagnostic tools to detect TET2 function in patients.

Lack of Major Genome-Wide DNA Methylation Changes in Succinate-Treated Human Epithelial Cells.

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The tricarboxylic acid (TCA) metabolite, succinate, is a competitive inhibitor of dioxygenase enzymes that require alpha ketoglutarate as a cofactor. One family of dioxygenases are the ten-eleven translocation (TET) proteins, which oxidize 5-methylcytosine to promote DNA demethylation. Inhibition of DNA demethylation is expected to lead to DNA hypermethylation, at least at genomic regions at which TET proteins are engaged. We treated human bronchial epithelial cells with succinate for five days and confirmed its effect on TET protein function by observing diminished formation of 5-hydroxymethylcytosine, the first oxidation product of the TET enzymatic reaction. We then analyzed global DNA methylation patterns by performing whole-genome bisulfite sequencing. Unexpectedly, we did not observe differentially methylated regions (DMRs) that reached genome-wide statistical significance. We observed a few regions of clustered DNA hypomethylation, which was also not expected based on the proposed mechanisms. We discuss potential explanations for our observations and the implications of these findings for tumorigenesis.

The level of active DNA demethylation compounds in leukocytes and urine samples as potential epigenetic biomarkers in breast cancer patients.

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The active DNA demethylation process, which involves TET proteins, can affect DNA methylation pattern.

TET dependent demethylation results in DNA hypomethylation by oxidation 5-methylcytosine (5-mC) to 5-hydroxymethylcytosine (5-hmC) and its derivatives. Moreover, TETs' activity may be upregulated by ascorbate. Given that aberrant DNA methylation of genes implicated in breast carcinogenesis may be involved in tumor progression, we wanted to determine whether breast cancer patients exert changes in the active DNA demethylation process. The study included blood samples from breast cancer patients (n = 74) and healthy subjects (n = 71). We analyzed the expression of genes involved in the active demethylation process (qRT-PCR), and 5-mC and its derivatives level (2D-UPLC MS/MS). The ascorbate level was determined using UPLC-MS. Breast cancer patients had significantly higher TET3 expression level, lower 5-mC and 5-hmC DNA levels. TET3 was significantly increased in luminal B breast cancer patients with expression of hormone receptors. Moreover, the ascorbate level in the plasma of breast cancer patients was decreased with the accompanying increase of sodium-dependent vitamin C transporters (SLC23A1 and SLC23A2). The presented study indicates the role of TET3 in DNA demethylation in breast carcinogenesis.
